Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

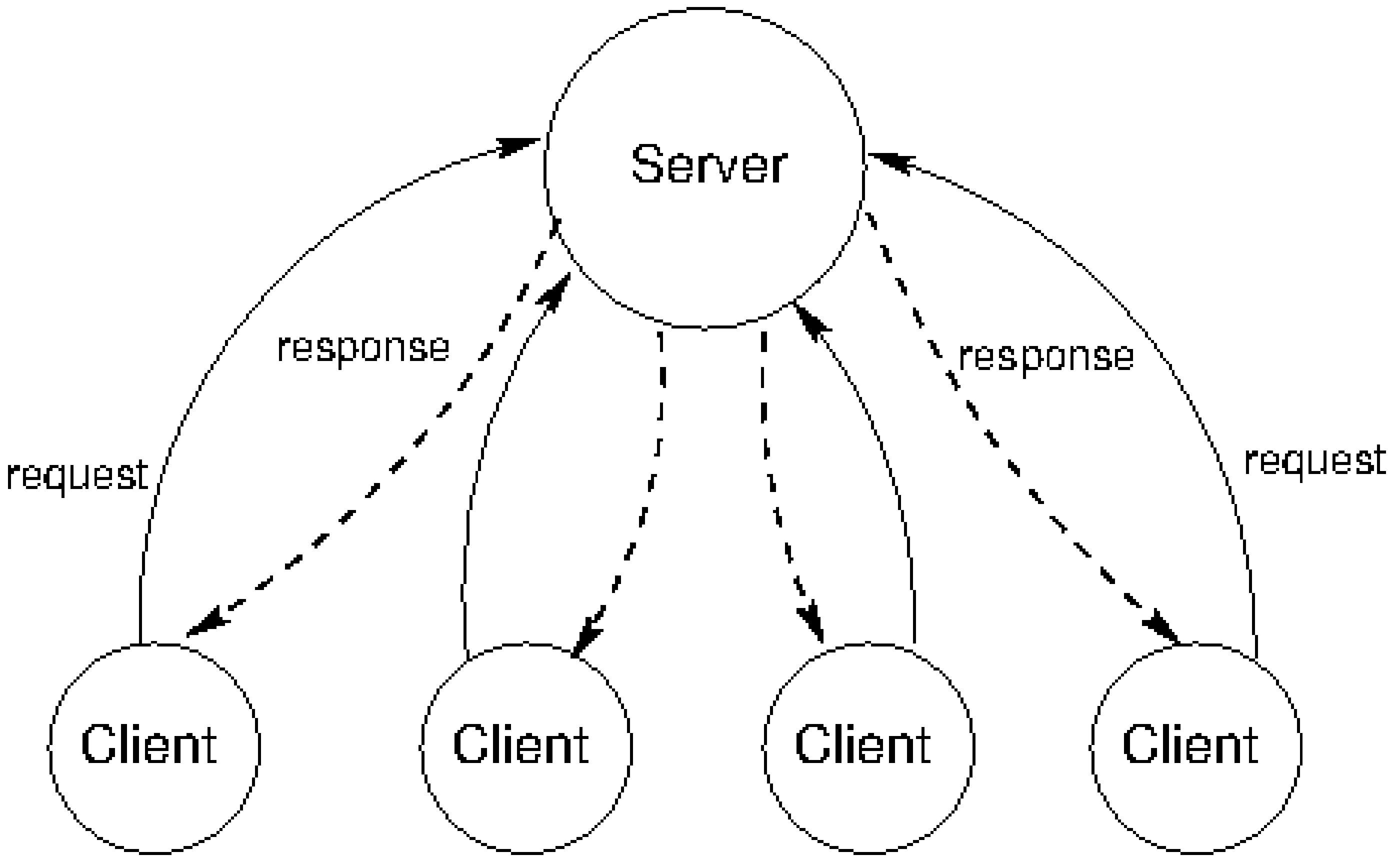
SSL Certificates *for* Security HTTPS Connections
(*Secure Sockets Layer*) = (HTTP *to* HTTPS)

Welcome to *the* Back-web Development *course*

1 - TLS/SSL Implementation (HTTP *to* HTTPS)

Web Client & Web Server Communication

A **<u>web client</u>** is an application that communicates with a web server, using Hypertext Transfer Protocol (HTTP). The basic objective of the web server is to store, process and deliver web pages to the users. **<u>Web browser is an example of a web client</u>**. Basically, a **<u>web server is a software that runs continuously and responds to requests</u>** by other pieces of software, usually called a client.



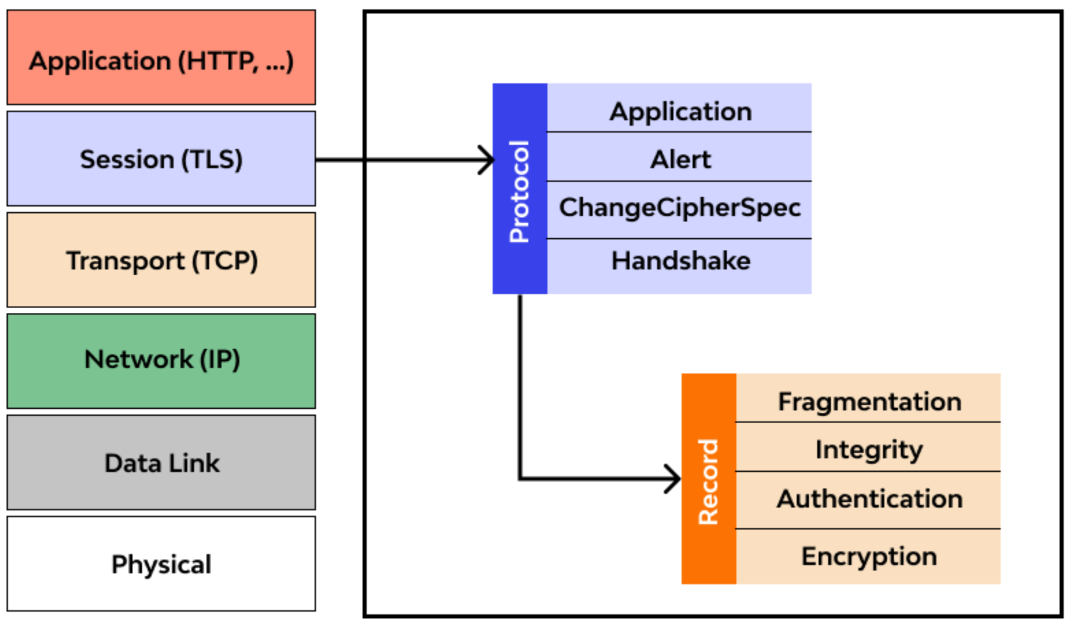
Security in the context of HTTP and HTTPS protocols, will inevitably involve the use of TLS (Transport Layer Security) and SSL (Secure Sockets Layer). The **implementation of TLS/SSL (HTTP to HTTPS) is essential to improve the security of web communication (client and server)**, especially when sensitive or private data must be transferred between the browser and the server, such as *login information*, *credit card details*, *customer data*, *government data* and others.

TLS (Transport Layer Security)

- Transport Layer Security (TLS) is a **cryptographic protocol designed to provide communications security** over a computer network. The protocol is widely used in applications such as email, instant messaging, and voice over IP, but its use in securing HTTPS remains the most publicly visible.

- TLS in web communication is a security protocol that **provides encryption and authenticity of data transferred over a network**. TLS replaces SSL and is designed to provide an additional layer of security on top of transport protocols such as TCP (Transmission Control Protocol).

- **TLS in web communication ensures that the data transferred between the user device and the server remains private and cannot be modified by unauthorized parties**. It involves the process of exchanging encryption keys and establishing a security session.

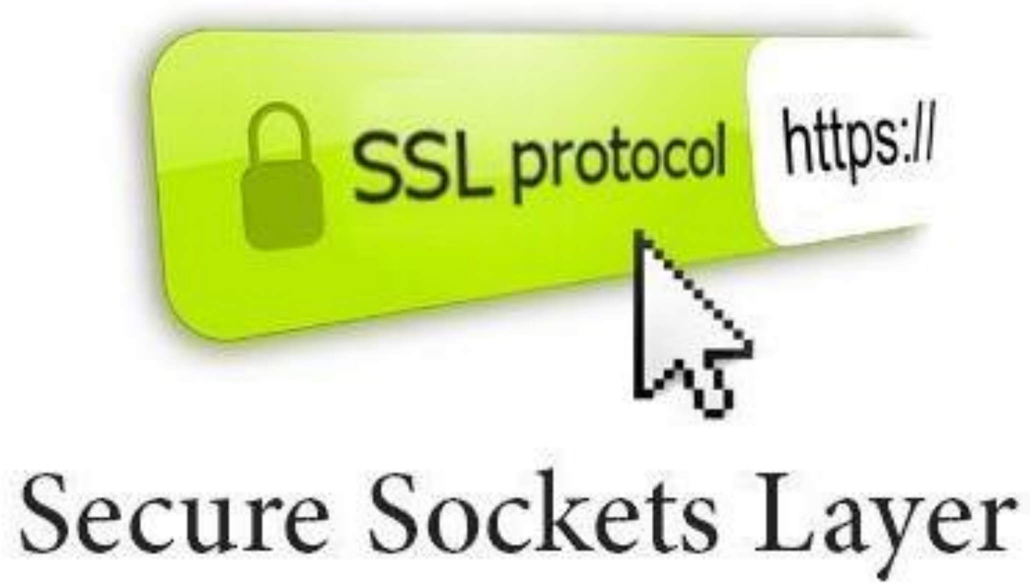


SSL (Secure Sockets Layer)

- SSL (Secure Sockets Layer) is an encryption-based Internet security protocol. It was first developed by Netscape in 1995 for the purpose of ensuring privacy, authentication, and data integrity in Internet communications.

- SSL is the predecessor **(pendahulu TLS)** to the modern TLS encryption used today. SSL in web communications was **the predecessor of TLS** and was used to provide similar security in transferring data over a network.

- SSL in web communications uses symmetric and asymmetric cryptography to protect data and provide trust in the identity of the server and, in some cases, in the client.



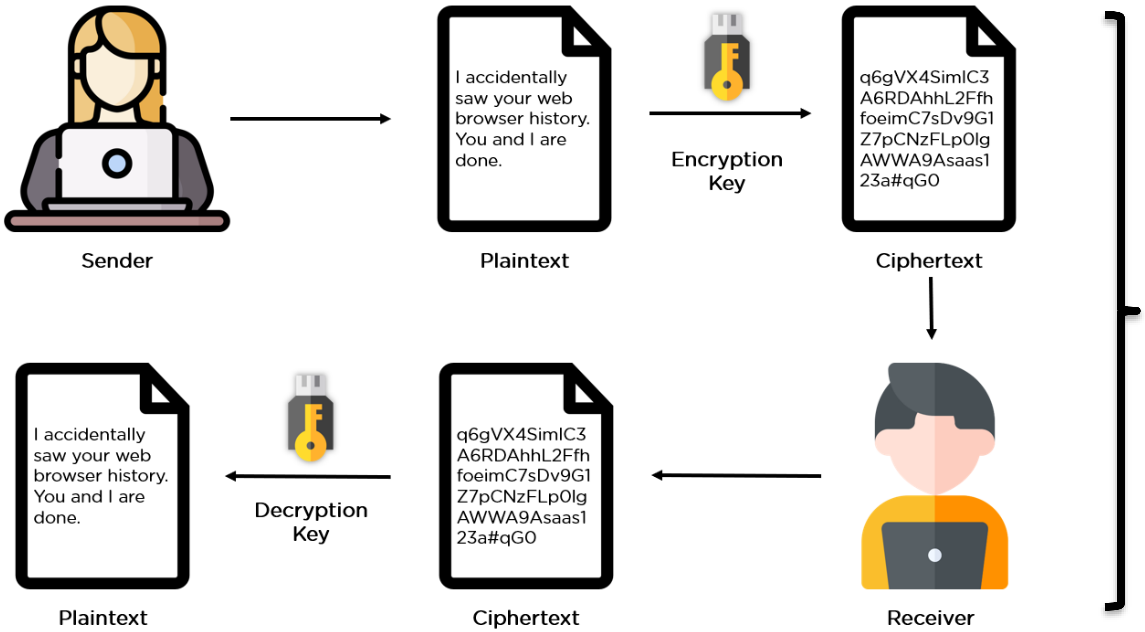
HTTP *to* HTTPS

HTTP (Hypertext Transfer Protocol) is a protocol used to transfer data on the web. However, HTTP does not provide adequate security (*tidak menyediakan keamanan yang memadai*) because the **data transmitted between the web browser and the server is not encrypted**, making it vulnerable to attacks and monitoring by third parties.

HTTPS (Hypertext Transfer Protocol Secure) is a secure version of HTTP. In HTTPS, the **data transferred between the web browser and the server is encrypted** using a security protocol, such as TLS or SSL, to protect the confidentiality and **integrity of the data**.

Mengapa komunikasi ini perlu dijaga?

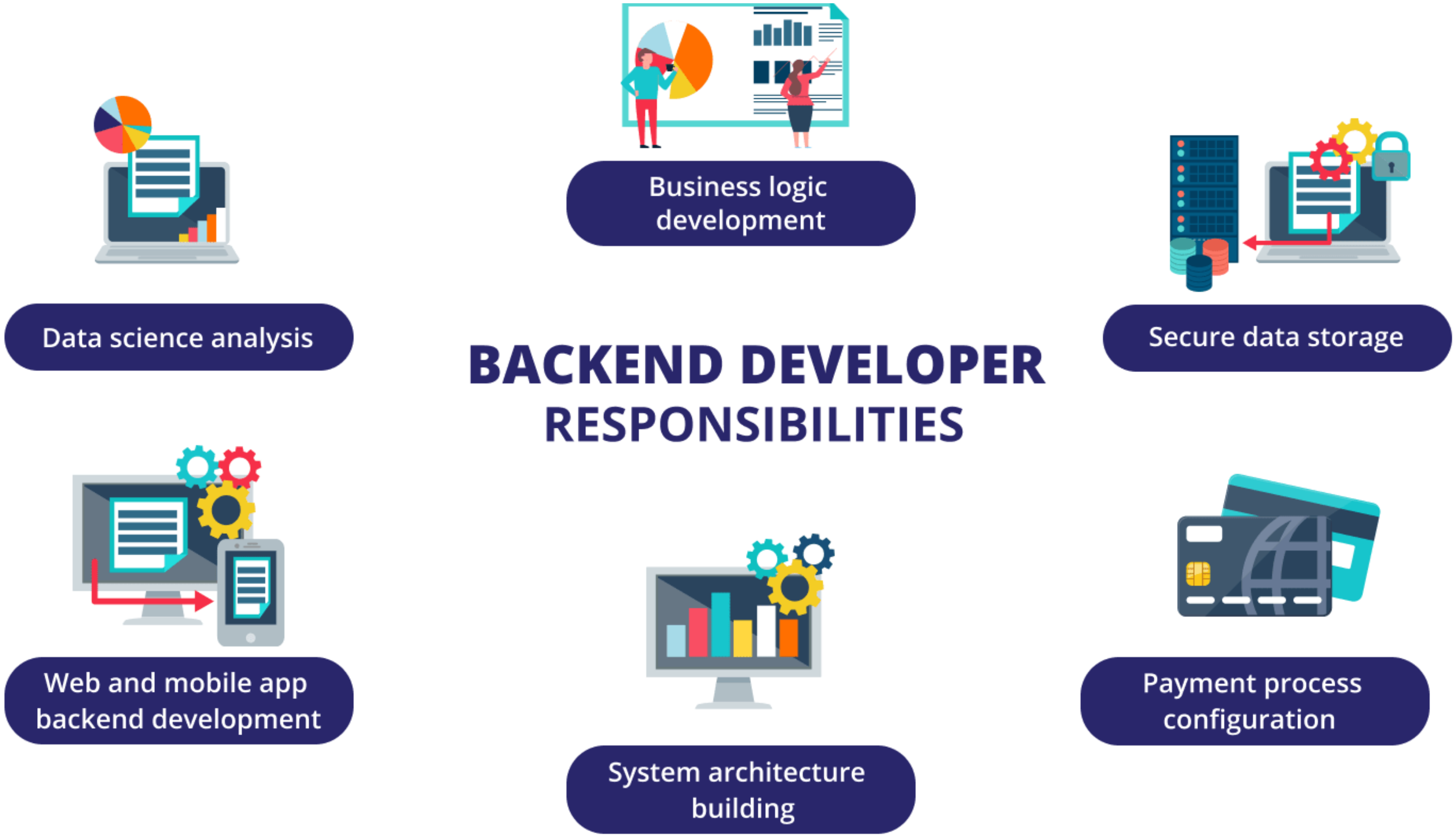
Many websites today, especially those that provide ***financial services***, ***e-commerce***, or ***collect personal information***, invest substantial resources into maintaining the security. These systems will undoubtedly use security protocols such as HTTPS to encrypt communication between users and servers, implement firewalls and intrusion detection systems, and regularly conduct security audits.



"The Internet is a dangerous place"

Internet dapat menjadi lingkungan yang berisiko karena adanya potensi ancaman keamanan seperti malware, phishing, and other cyber crimes. Selain itu, internet juga dapat menjadi tempat di mana informasi pribadi dapat terancam keamanannya (compromised).

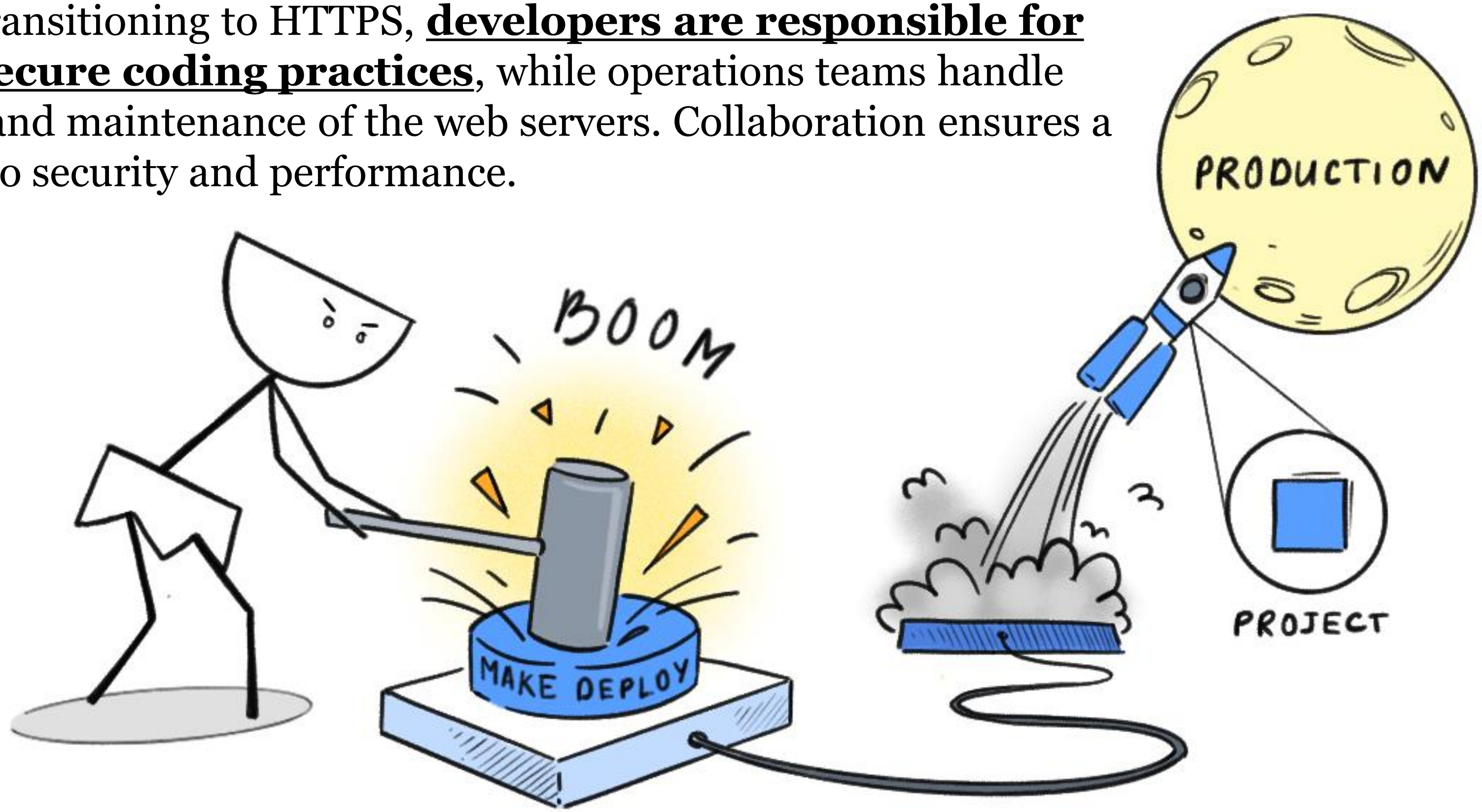
2 – Why “HTTP *to* HTTPS” important *for* Back-end developer?



It is important for **<u>web developers</u>** on both the **front-end and back-end sides** to understand the concepts of HTTP and HTTPS as well as SSL certificates because these are related to security and data integrity on the web. By understanding the concepts of HTTP, HTTPS, and SSL certificates, web developers (*website and server side*) can optimize security and user confidence in the website being built. This is also in line with industry trends and evolving security requirements.

Collaboration between Dev & Ops

DevOps encourages collaboration between development and operations teams. In the context of transitioning to HTTPS, **developers are responsible for implementing secure coding practices**, while operations teams handle the configuration and maintenance of the web servers. Collaboration ensures a holistic approach to security and performance.

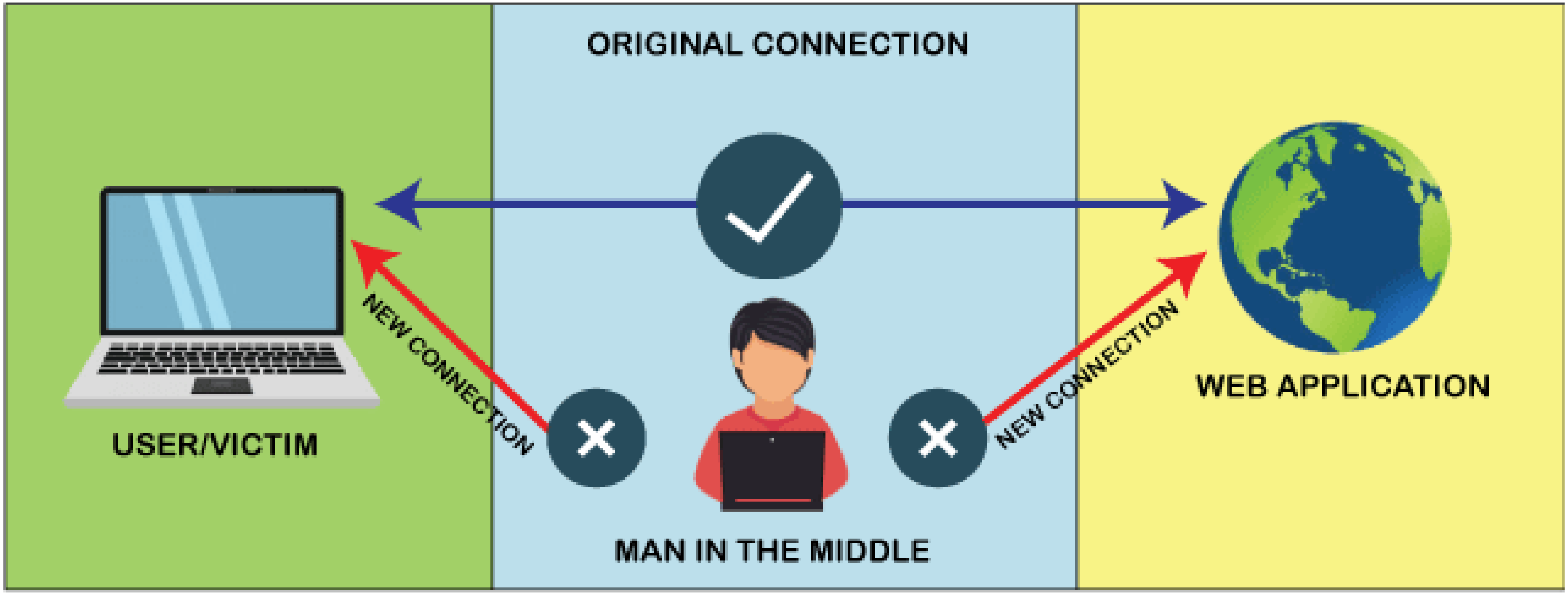


DevOps practices often **emphasize security as an integral part of the development lifecycle**. Transitioning from HTTP to HTTPS is a security measure to encrypt data in transit, providing a secure communication channel between the web client and server. DevOps teams may be involved in implementing and managing the HTTPS configuration to ensure compliance with security standards.

User Data Security

- HTTPS (Hypertext Transfer Protocol Secure) uses SSL/TLS (Secure Sockets Layer/Transport Layer Security) to **encrypt the data sent between the server and the user**. Thus, sensitive information such as passwords, credit card information, and personal data of the user remains safe from attacks by hackers trying to steal or manipulate the data.
- Modern users tend to be more trusting of websites that use HTTPS. Modern browsers often mark HTTP sites as insecure, while HTTPS sites are given a mark indicating that the connection is secure. This can increase user trust in your website.
- Search engines, such as Google, give higher rankings to websites that use HTTPS. Therefore, using an SSL certificate and building your website with the HTTPS protocol can provide advantages in terms of SEO.
- SSL certificates help **prevent man-in-the-middle attacks**, where an **attacker tries to spy** on or alter the data sent between the user and the server.

Data Integrity (No Data manipulation)

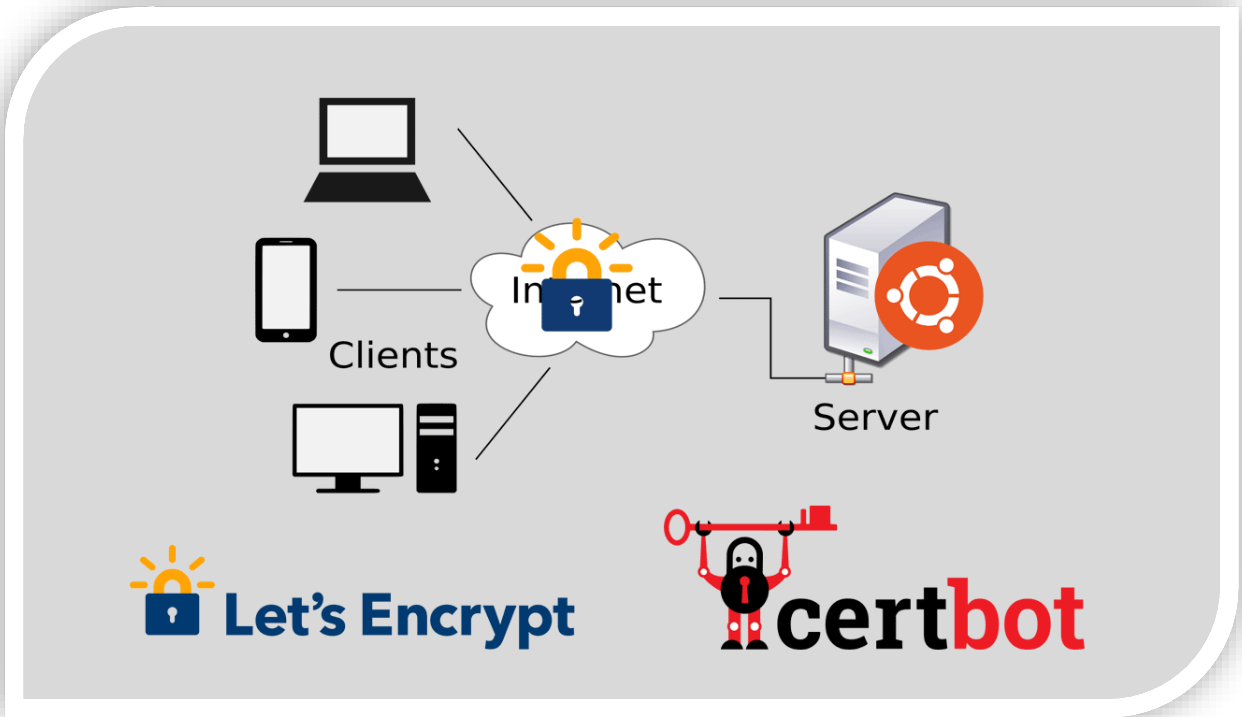


- **Man in the Middle (MiTM)** is a *cyberattack* carried out to steal information and spy on the victim. MiTM perpetrators (Pelaku MiTM) will **<u>take advantage of insecure internet connections</u>** to carry out their actions. MiTM will position itself between the victim and the website being used. Without realizing the victim is being observed by hackers.

- SSL certificates (HTTP to HTTPS connections) not only encrypt data, but also provide data integrity. This means that the data sent and received cannot be manipulated by third parties during the transmission process.

SSL certificates (Secure Sockets Layer) are what enable websites to use HTTPS, which is more secure than HTTP. An SSL certificate is a data file hosted in a website's origin server. SSL certificates make SSL/TLS encryption possible, and they contain the website's public key and the website's identity, along with related information.

Let’s Encrypt SSL Certificate



Let's Encrypt is a free, automated, and open certificate authority. Let's Encrypt is a global Certificate Authority (CA) to let people and organizations around the world obtain, renew, and manage SSL/TLS certificates. These certificates can be used by websites to enable secure HTTPS connections. Let's Encrypt offers Domain Validation (DV) certificates.

Common Requirement SSL Certificates

1) Requires a **DOMAIN NAME** to secure with an SSL certificate. Let's Encrypt provides certificates for public domain names that are accessible over the internet.

2) Requires access to the **SERVER** where the domain is hosted. This can be SSH access or any other type of access that allows us to manage the server configuration.

3) Need to have a **WEB SERVER** running on the server. Apache, Nginx, and several other web servers are supported by Let's Encrypt.

4) Need to have **CERTBOT (ACME CLIENT)** or another ACME (Automatic Certificate Management Environment) client installed on the server. Certbot is the official tool supported by **<u>Let's Encrypt to automate SSL certificate management</u>**. Ensure that the server's operating system supports Certbot and the latest version of Let's Encrypt. Ubuntu 22.04 should support Certbot well.

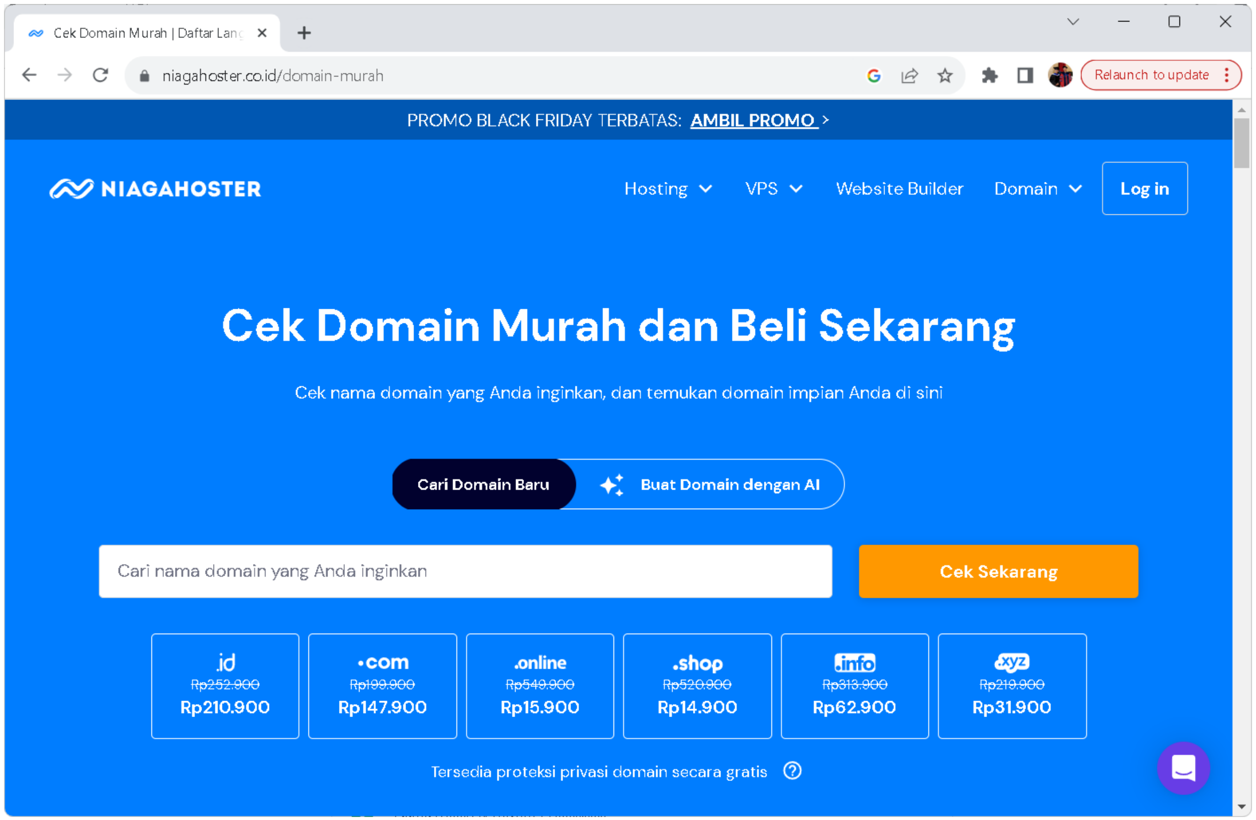
Exercise *for* Students

Exercise #1

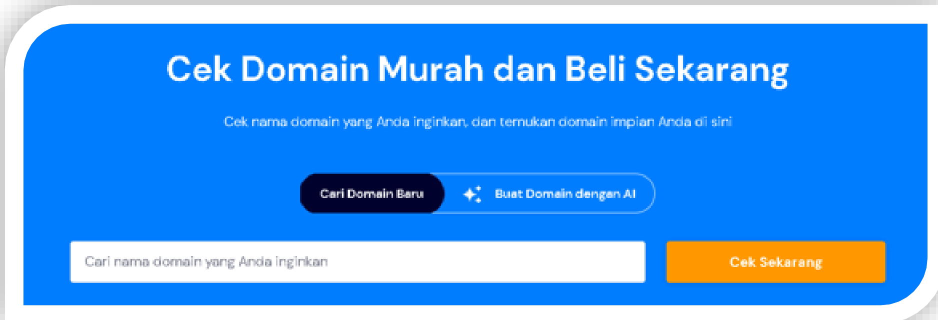
(Buat domain name murah di niagahoster.co.id)

Open & Sign In NiagaHoster

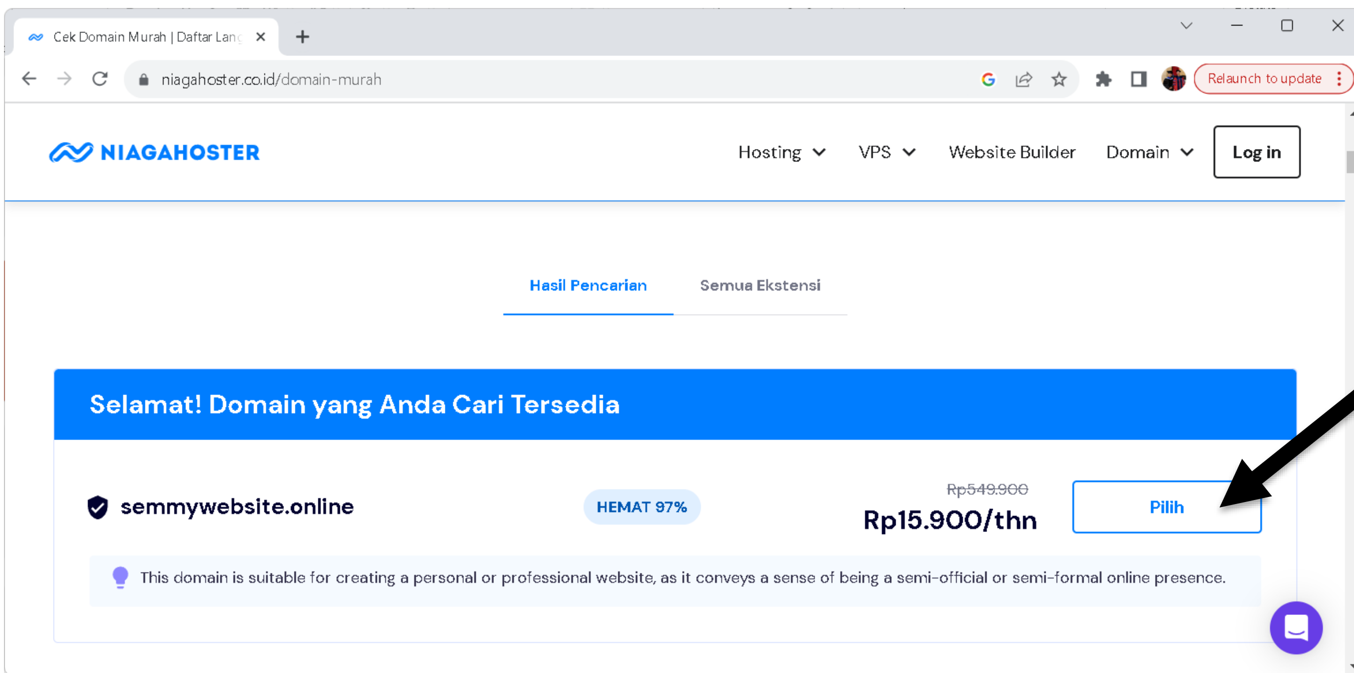
https://www.niagahoster.co.id/domain-murah



Cari Domain Murah

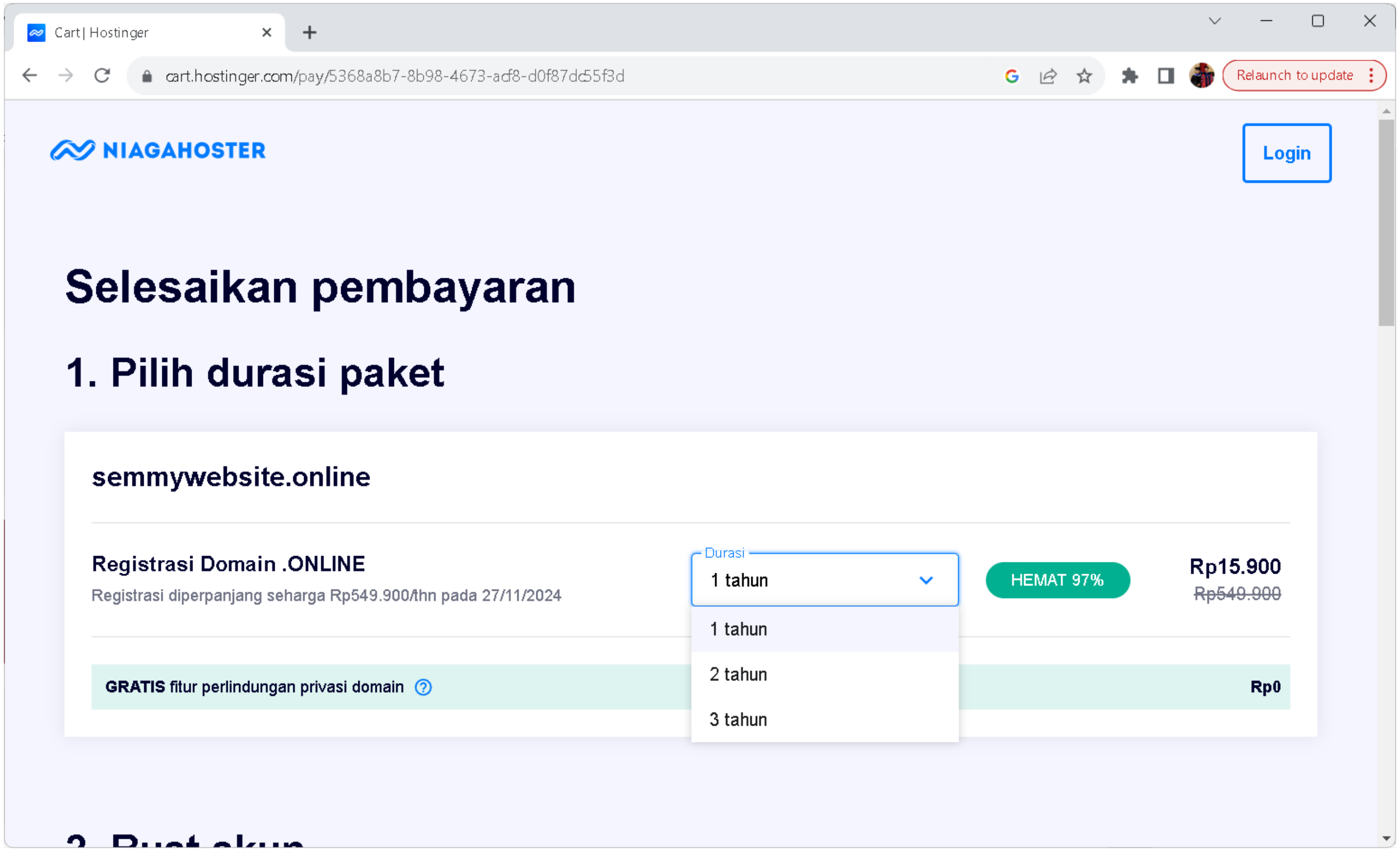


Cari "semmywebsite.online"

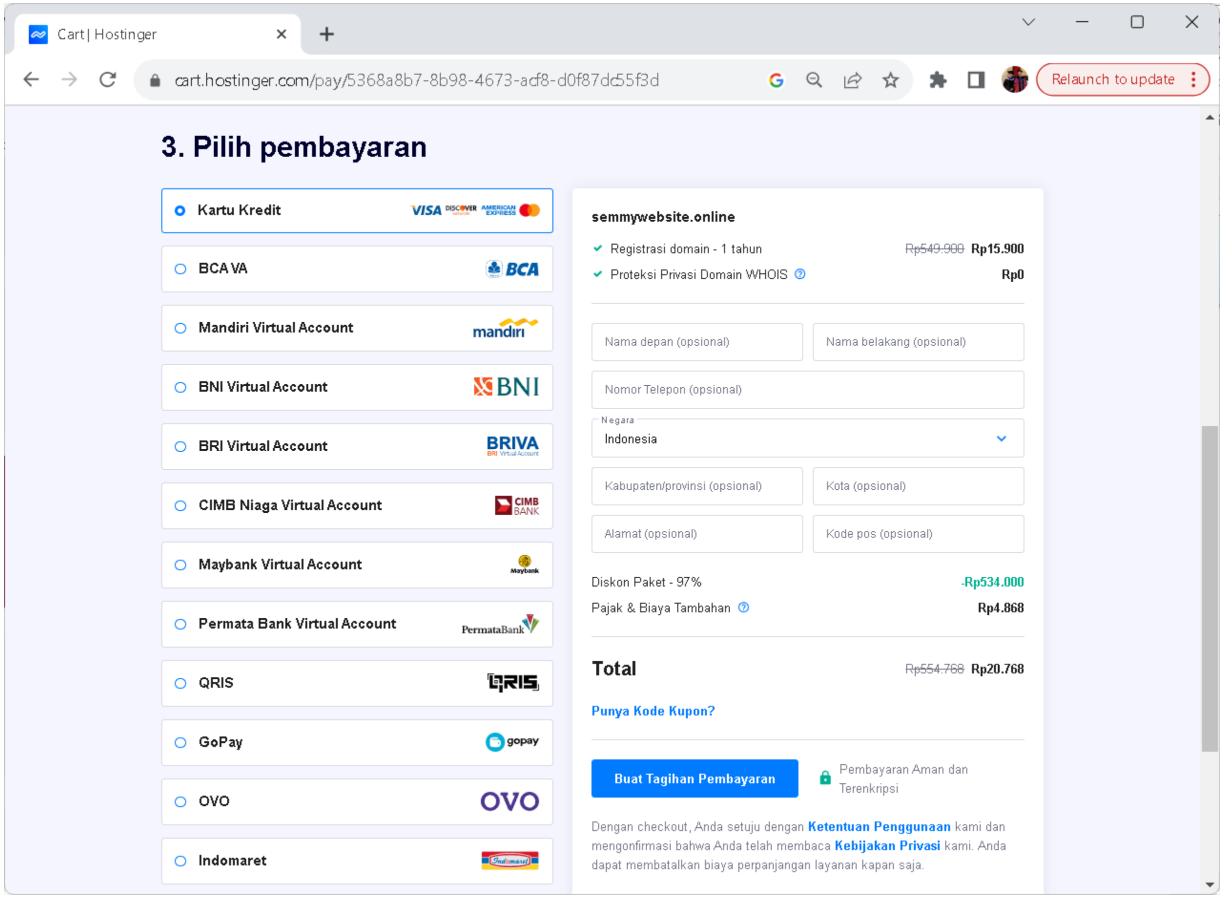


Click button ini untuk memilih domain name.

Durasi Registrasi Domain

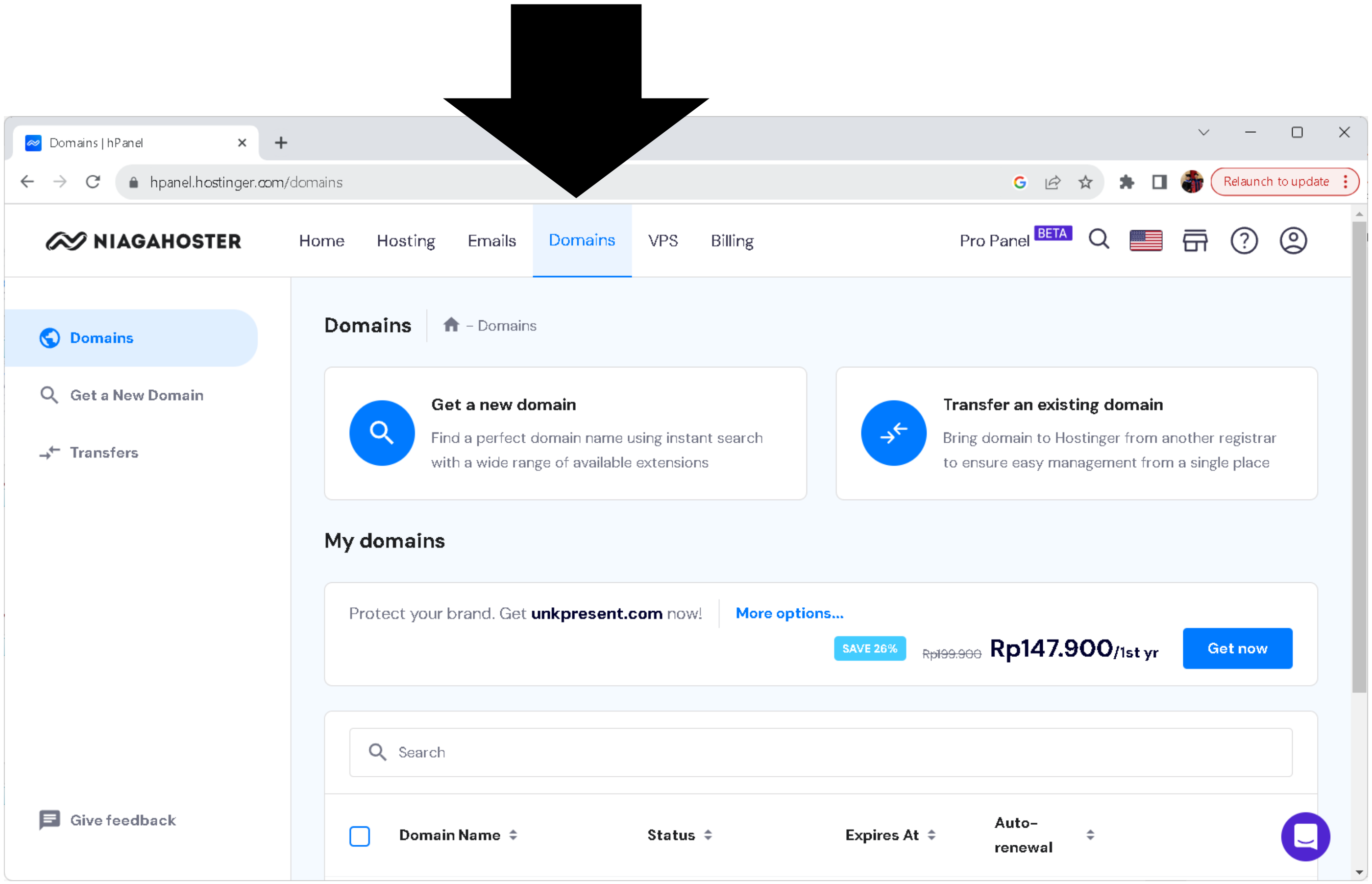


Metode Pembayaran

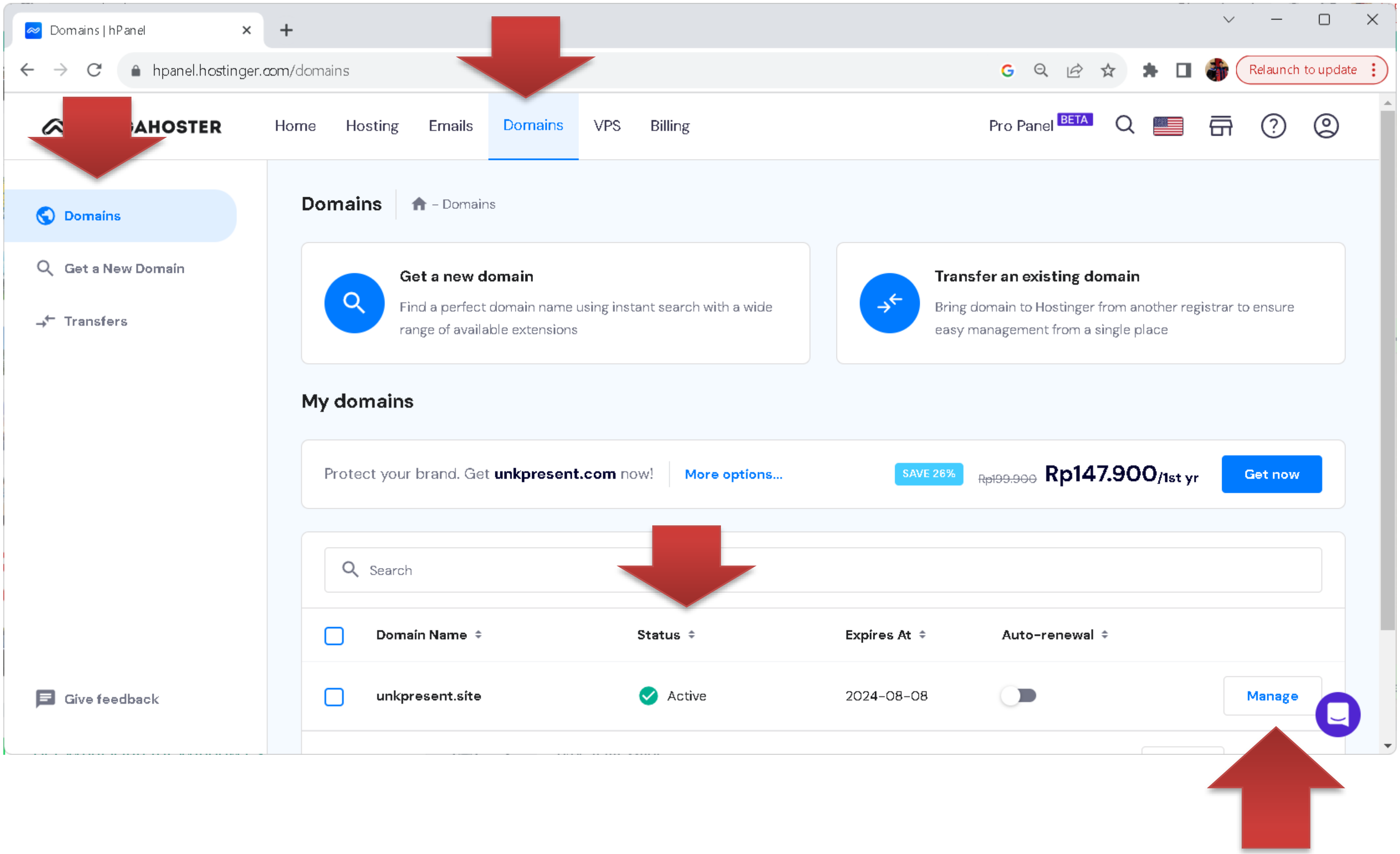


Masuk Ke Dashboard Niagahoster

Click di tab Domains

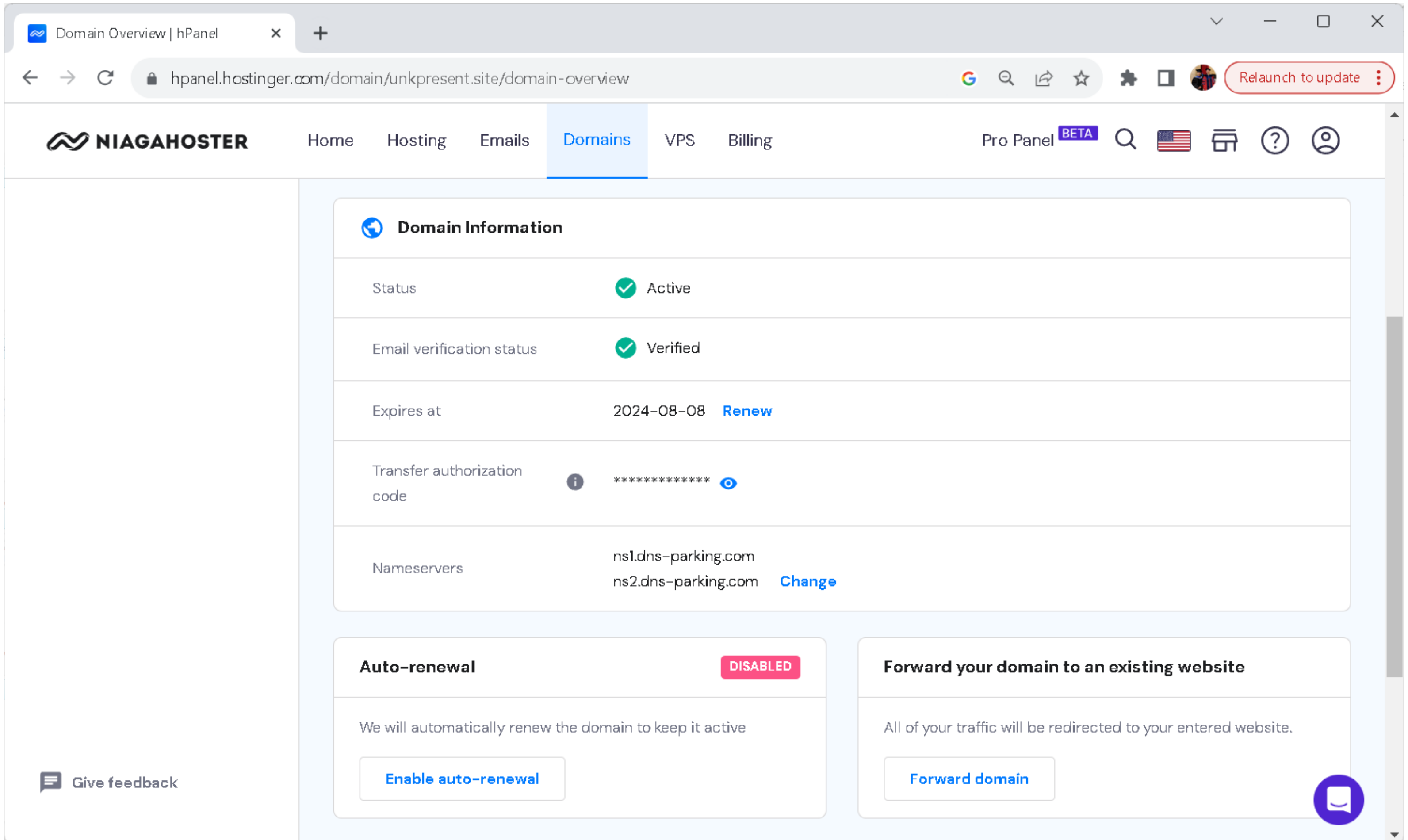


Cek Status Domain Name

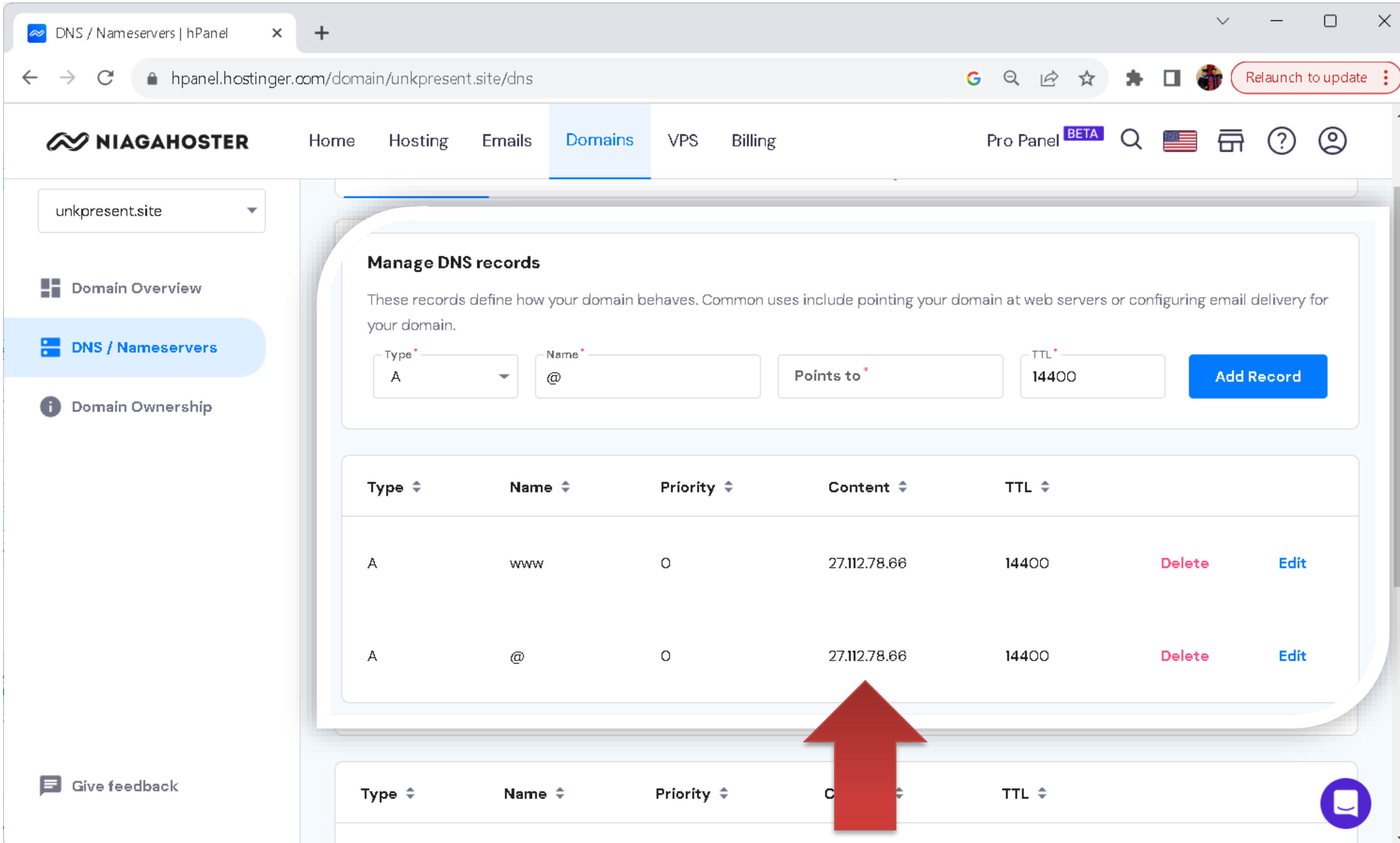


Click button “Manage”

Pastikan Sudah Verified by Emain



Click Tab “DNS/Nameservers”



Change with your public IP Address

Exercise #2

(Install Let's Encrypt SSL Certificate on Ubuntu 22.04)

Install Certbot Client

1) Updates the package lists:
- **sudo apt-get update**

2) Install the **Certbot client** with the Apache plugin:
- **sudo snap install --classic certbot**
- sudo apt-get install certbot python3-certbot-apache
- sudo apt-get install certbot --fix-missing
- sudo apt-get install certbot python3-certbot-apache apache2

3) Test if the Certbot client was installed correctly:
- **sudo certbot --help**

4) Obtain certificate for domain name:
- **sudo certbot --apache**

5) Add (Multiple) New domain name:
- **sudo certbot --apache -d presensi.unklab.ac.id**

6) Delete old certificate (sertifikat ssl lama):
- **sudo certbot delete**

Sesudah perintah ini dijalankan, anda akan diminta untuk memasukan email dan domain name yang telah dibuat.

Renew Certificate (after 3 months)

1) Certbot is on Autorenewal:

- sudo systemctl list-timers
- sudo systemctl enable certbot.timer
- sudo systemctl start certbot.timer
- sudo systemctl status certbot.timer
- sudo certbot renew --dry-run

2) Cek Status Certificate:

- sudo certbot certificates

3) Renew Certificate:

- sudo certbot renew

Check Configuration 443

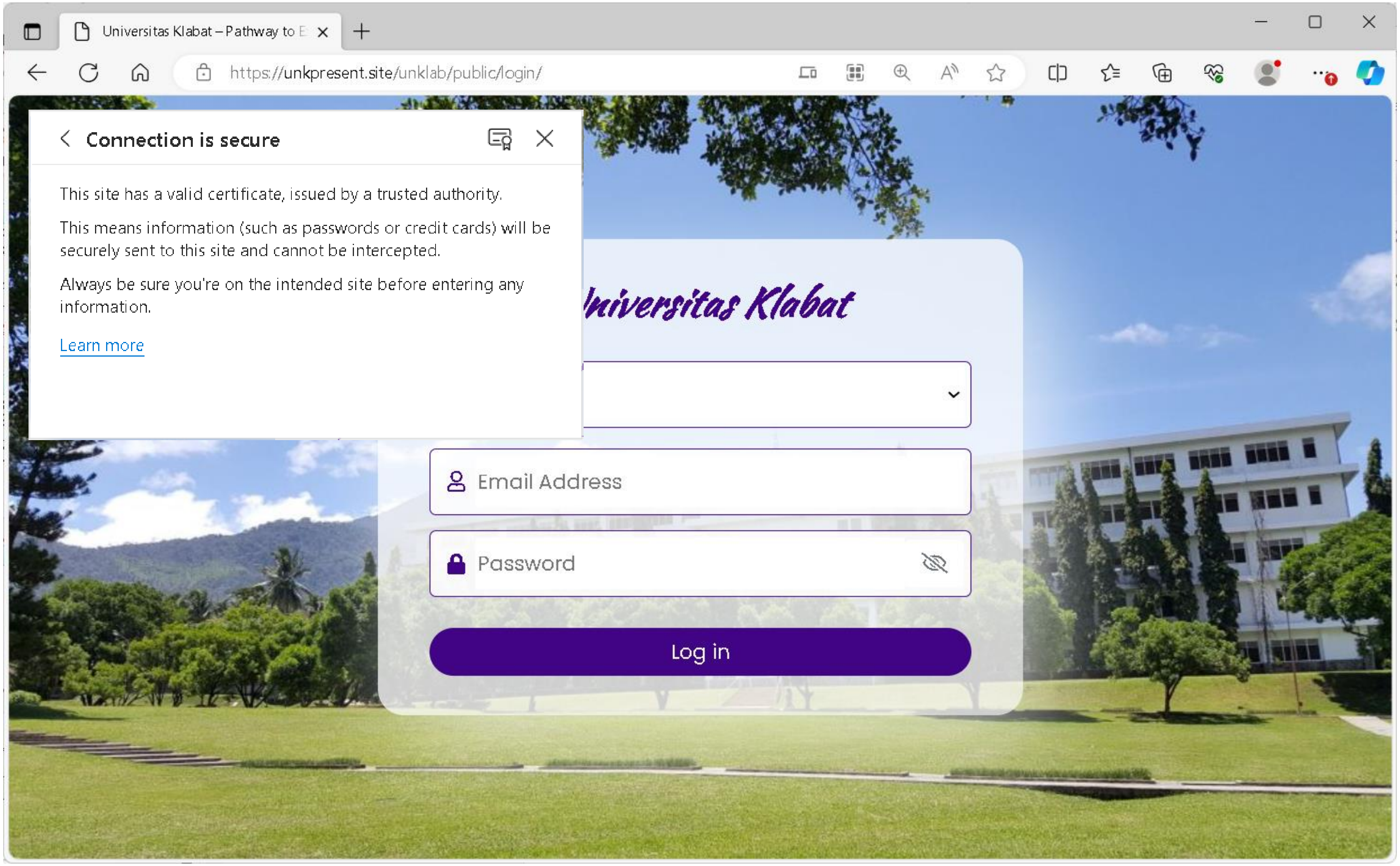
Certbot is on Autorenewal:

- sudo nano /etc/apache2/sites-enabled/000-default-le-ssl.conf

root@UnklabPresensi: /home,

```
  GNU nano 6.2                    /etc/apache2/sites-enabled/000-default-le-ssl.conf
<IfModule mod_ssl.c>
<VirtualHost *:443>
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        #ServerName www.example.com

        ServerAdmin webmaster@localhost
        DocumentRoot /var/www/html

        # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn

        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined

        # For most configuration files from conf-available/, which are
        # enabled or disabled at a global level, it is possible to
        # include a line for only one particular virtual host. For example the
        # following line enables the CGI configuration for this host only
        # after it has been globally disabled with "a2disconf".
        #Include conf-available/serve-cgi-bin.conf


ServerName presensi.unklab.ac.id
SSLCertificateFile /etc/letsencrypt/live/presensi.unklab.ac.id/fullchain.pem
SSLCertificateKeyFile /etc/letsencrypt/live/presensi.unklab.ac.id/privkey.pem
Include /etc/letsencrypt/options-ssl-apache.conf
</VirtualHost>
</IfModule>



^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location    M-U Undo       M-A Set Mark
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo       M-6 Copy
```

Check Https Access



Exercise #3

(Redirect Public IP (80 HTTP) ke Domain (443 HTTPS))

Check File Apache Config Aktif

Tujuannya adalah supaya akses lewat IP (contoh: http://103.xxx.xxx.xxx) otomatis pindah ke domain dengan HTTPS, berikut adalah step-by-step di server Apache2:

#1 Cara cek file config aktif. Jalankan:

sudo apache2ctl -S

```
root@ISC2025:/home/iscadmin# sudo apache2ctl -S
AH00558: apache2: Could not reliably determine the server's fully qualified
domain name, using 10.50.189.59. Set the 'ServerName' directive globally to
suppress this message
VirtualHost configuration:
*:443                  isc.unklab.ac.id (/etc/apache2/sites-enabled/000-defa
ult-le-ssl.conf:2)
*:80                   10.50.189.59 (/etc/apache2/sites-enabled/000-default.
conf:1)
ServerRoot: "/etc/apache2"
Main DocumentRoot: "/var/www/html"
Main ErrorLog: "/var/log/apache2/error.log"
Mutex watchdog-callback: using_defaults
Mutex rewrite-map: using_defaults
Mutex ssl-stapling-refresh: using_defaults
Mutex ssl-stapling: using_defaults
Mutex ssl-cache: using_defaults
Mutex default: dir="/var/run/apache2/" mechanism=default
Mutex mpm-accept: using_defaults
PidFile: "/var/run/apache2/apache2.pid"
Define: DUMP_VHOSTS
```

Redirect Permanent

#2 Edit file 000-default.conf (default file ini saja sudah cukup):

sudo nano /etc/apache2/sites-available/000-default.conf

#3 Isi jadi seperti ini (Ganti public IP address di bawah ini):

```
<VirtualHost *:80>
   ServerAdmin webmaster@localhost
   ServerName 113.193.158.86
   ServerAlias isc.unklab.ac.id www.isc.unklab.ac.id

   Redirect permanent / https://isc.unklab.ac.id/

   ErrorLog ${APACHE_LOG_DIR}/error.log
   CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>
```

Dengan begini, semua request ke port 80 (baik pakai IP maupun domain) langsung di-redirect ke https://isc.unklab.ac.id/.

Reload Apache

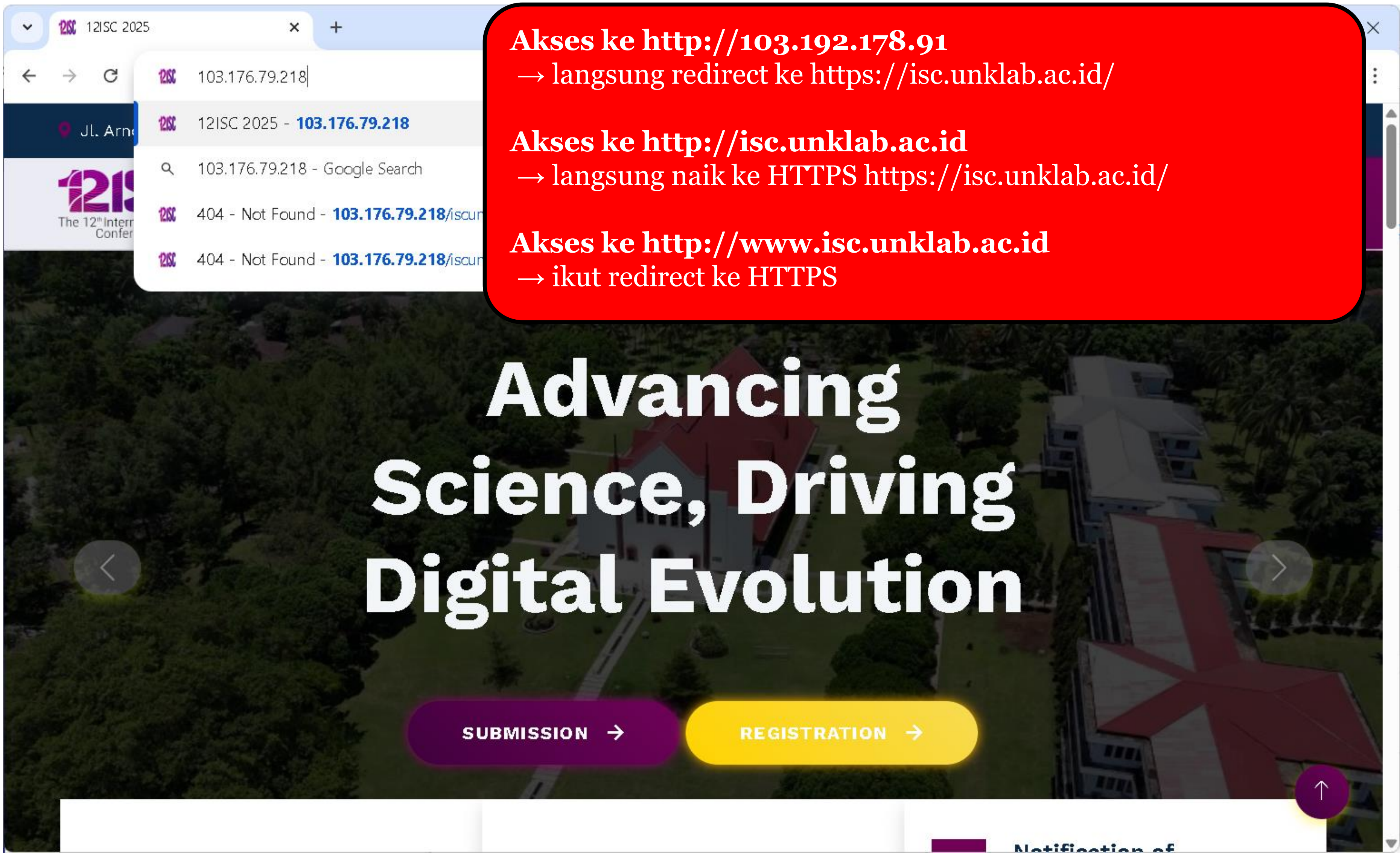
#4 Simpan lalu reload Apache:
sudo systemctl reload apache2

(<u>Opsional</u>) Tambahkan config khusus supaya semua request selain domain diarahkan. Kalau kita ingin **semua host/ip apapun yang diarahkan ke server ini** tetap dialihkan ke domain utama, bisa pakai:

```
<VirtualHost *:80>
   ServerName default
   Redirect permanent / https://isc.unklab.ac.id/
</VirtualHost>
```

#5 Reload Apache
sudo systemctl reload apache2

Output



Akses ke http://103.192.178.91
→ langsung redirect ke https://isc.unklab.ac.id/

Akses ke http://isc.unklab.ac.id
→ langsung naik ke HTTPS https://isc.unklab.ac.id/

Akses ke http://www.isc.unklab.ac.id
→ ikut redirect ke HTTPS

END PRESENTATION

Thank you for your attention

Instructor: S – W – T